

AMAZON ORDERS DATA ANALYSIS REPORT

Using Python, Excel, SQL, and Power BI

INTRODUCTION:-

The project focuses on analyzing Amazon orders data using Python, Excel, SQL and Power BI to identify meaningful business insights related to sales trends, customer behavior, revenue generation, shipping performance, promotions, and cancellations.

The dataset contains information such as order details, product categories, sizes, order status, ship-city, ship-state, quantity, and revenue amount.

The main objective of this project is to clean, analyze, visualize, and interpret the data to help Amazon sellers and entrepreneurs make data-driven business decisions.

The project also includes the development of an interactive Power BI dashboard for effective business reporting and visualization.

TOOLS USED:-

<u>Tool</u>	<u>Purpose</u>
Python	Data Cleaning & Preprocessing
Excel	Pivot Tables & Initial Analysis
SQL	Aggregations & Analytical Queries
Power BI	Interactive Dashboard & KPI Visualization

DATA CLEANING & PREPROCESSING:-

Before analysis, the dataset required extensive cleaning and preprocessing to ensure data quality and accuracy. Python was used to clean and standardize the dataset.

The following operations were performed:

- Removed duplicate records
- Handled missing/null values
- Standardized categorical columns
- Converted date and numeric columns

- Corrected inconsistent formatting
- Removed unnecessary spaces and invalid entries
- Created derived columns for analysis

These preprocessing steps improved data reliability and ensured accurate analysis and visualization.

EXPLORATORY DATA ANALYSIS (EDA):-

EDA was performed using Python and Excel to understand the structure and distribution of the dataset.

The following analyses were conducted:

- Product category distribution
- Product size distribution
- Order status distribution
- Orders by city and state
- Revenue trend analysis
- Shipping service analysis
- Promotion impact analysis
- Cancellation trend analysis

The analysis helped identify customer purchasing behavior, high-performing regions, and sales patterns.

SQL ANALYSIS:-

SQL was used to perform aggregation-based and grouped analysis on the Amazon orders dataset. Various SQL queries were written to analyze:

- Top-performing states and cities
- Revenue trends
- Orders by category
- Shipping service analysis
- Fulfillment channel performance
- Promotion impact on sales
- Cancellation analysis by category and size

SQL helped efficiently process large volumes of transactional data and generate meaningful business insights.

POWER BI DASHBOARD OVERVIEW:-

An interactive multi-page Power BI dashboard was created to visualize key business insights and KPIs from the Amazon orders dataset. The dashboard includes:

- Overview KPI metrics
- Orders analysis
- Revenue analysis
- Promotion and cancellation analysis
- Shipping service analysis
- Detailed orders explorer

Interactive visuals, charts, slicers, and filters were used to improve data exploration and business decision-making.

KEY BUSINESS INSIGHTS:-

Overall Business Performance

- Total Orders Analyzed = **128,975**
- Total Revenue Generated = **83.64 Million**
- Average order Amount = **648.51**

1. Top Performing Categories:

The “**Set**” category generated the highest revenue with approximately **41.3 Million**, followed by “**Kurta**” with **23.3 Million**.

<u>Category</u>	<u>Revenue</u>
Set	41.3M
Kurta	23.3M
Western Dress	11.7M
Top	5.6M

These categories contribute the majority of sales and should receive higher inventory and marketing focus.

2. Category Distribution:-

The majority of orders belong to a few major categories.

<u>Category</u>	<u>Order Contribution</u>
Set	39%
Kurta	38.7%
Western Dress	12%
Top	8.2%

This shows strong customer demand for ethnic and apparel products.

3. Top Revenue Generating States:

<u>State</u>	<u>Revenue</u>
Maharashtra	14.1M
Karnataka	11.1M
Telangana	7.4M
Uttar Pradesh	7.3M
Tamil Nadu	7M

A few states contribute a major share of total revenue, indicating stronger market penetration in these regions.

4. Top Revenue Generating Cities:

<u>City</u>	<u>Revenue</u>
Bengaluru	7.6M
Hyderabad	5.9M
Mumbai	4.5M
New Delhi	4.2M
Chennai	3.8M

Metro cities dominate customer purchases and revenue generation.

5. Order Status Analysis:

<u>Status</u>	<u>Percentage</u>
Shipped	60.3%
Delivered to Buyer	22.3%
Cancelled	14.2%

The cancellation rate of **14.2%** indicates the need for improvements in delivery, product descriptions, and size accuracy.

6. Product Size Analysis:

<u>Size</u>	<u>Order Percentage</u>
M	17.6%
L	17.2%
XL	16.2%
XXL	14%

Medium and Large sizes are the most preferred by customers.

6. Promotion Analysis:

Orders with promotions showed better revenue performance, indicating that discounts and offers positively influence customer purchasing behavior

7. Shipping Service Analysis:

Standard shipping services handled the majority of orders, while premium categories showed greater use of expedited shipping services.

FACTORS THAT INFLUENCING HIGHER ORDER AMOUNTS:-

The following factors contributed to higher order amounts:

- High-demand product categories
- Promotional offers and discounts
- Strong-performing metro cities
- Faster shipping services

- Higher quantity purchases
- Popular product sizes

BUSINESS STRATEGIES & RECOMMENDATIONS:-

▪ Inventory Optimization-

Focus inventory planning on high-performing categories such as “**Set**” and “**Kurta**” to avoid stock shortages.

▪ Regional Marketing Strategy-

Increase advertisements and promotional campaigns in high-performing cities such as Bengaluru, Hyderabad, and Mumbai.

▪ Reduce Cancellation Rates-

Improve:

- size charts
- product descriptions
- delivery timelines
- customer support
- to reduce the current cancellation rate.

▪ Promotion Strategy-

Use targeted discounts and festival offers to increase customer engagement and boost sales.

▪ Shipping Optimization-

Enhance logistics efficiency for frequently used shipping services to improve customer satisfaction.

RECOMMENDATIONS FOR NEW AMAZON SELLERS:-

New sellers should:

- Focus on high-demand categories
- Target metro cities initially
- Maintain competitive pricing
- Use promotional campaigns strategically
- Improve product images and descriptions
- Monitor cancellation patterns regularly

- Optimize shipping and delivery performance

These strategies can improve customer satisfaction, increase sales performance, and support long-term business growth.

CONCLUSION:-

This project successfully analyzed Amazon orders data using Python, Excel, SQL, and Power BI. The analysis identified important trends related to customer behavior, revenue generation, shipping performance, promotions, and cancellations.

The interactive Power BI dashboard and analytical insights generated through this project can help Amazon sellers make better data-driven decisions, improve operational efficiency, and increase overall profitability in the e-commerce marketplace.

